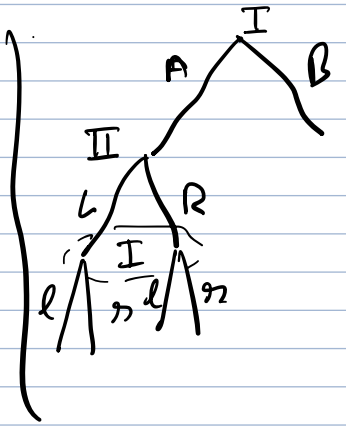
Class 29

Result In an extensive form game, every behavior strategy is outcome equivalent to a mixed strategy.

Proof Consider player i and a behavior strategy b_i.

Suppose player i has m many information sets $h_1, \ldots, h_m$.

$$\sigma_i(s_i) = \prod_{j=1}^{m} \beta_i(s_i(h_j), h_j) \quad \forall s_i \in S_i$$



Valied mixed strategy

as $\beta_i(s_i(h_j), h_j) \geq 0 \quad \forall j = 1, \ldots, m$

$\sigma_i(s_i) \geq 0$

and

$$\sum_{s_i \in S_i} \sigma_i(s_i) = \sum_{s_i \in S_i} \prod_{j=1}^{m} \beta_i(s_i(h_j), h_j)$$

$$= \prod_{j=1}^{m} \sum_{s_i \in S_i} \beta_i(s_i(h_j), h_j)$$

$$= \prod_{j=1}^{m} (1) \qquad \text{as } \beta_i \text{ is a behavior strategy}$$

$$= 1$$

σ_i is a valid mixed strategy.

Fix a leaf node z and suppose $h_1, \ldots, h_k$ are the information sets of player i on the path from root to z. Also, assume the $a_1, \ldots, a_k$ are the actions of player i on those information sets.

Under β_i player i's contribution to the probability of reaching z

$$\prod_{j=1}^{k} \beta_i(a_j, h_j)$$

Under σ_i, the contribution becomes

$$\sum_{\substack{s_i \in S_i \\ s_i(h_j) = a_j \\ \forall j = 1, \ldots, k}} \sigma_i(s_i)$$

$$= \sum_{\substack{s_i \in S_i \\ s_i(h_j) = a_j \\ \forall j = 1, \ldots, k}} \prod_{\ell=1}^{m} \beta_i(s_i(h_\ell), h_\ell)$$

$$= \sum_{\substack{s_i \in S_i \\ s_i(h_j) = a_j \\ \forall j = 1, \ldots, k}} \prod_{\ell=1}^{k} \beta_i(s_i(h_\ell), h_\ell) \prod_{\ell=k+1}^{m} \beta_i(s_i(h_\ell), h_\ell)$$

$$= \sum_{\substack{s_i \in S_i \\ s_i(h_j) = a_j}} \prod_{\ell=1}^{k} \beta_i(a_\ell, h_\ell) \prod_{\ell=k+1}^{m} \beta_i(s_i(h_\ell), h_\ell)$$

$\forall j = 1, \dots k$

$$= \prod_{l=1}^{k} \beta_i(a_l; h_l) \sum_{\substack{s_i \in S_i \\ s_i(h_j) = a_j \\ \forall j = 1, \dots k}} \prod_{l=k+1}^{m} \beta_i(s_i(h_l); h_l)$$

$$= \prod_{l=1}^{k} \beta_i(a_l; h_l) \sum_{s_i \in S_i} \prod_{l=k+1}^{m} \beta_i(s_i(h_l); h_l)$$

$$= \prod_{l=1}^{k} \beta_i(a_l; h_l) \prod_{l=k+1}^{m} \sum_{s_i \in S_i} \beta_i(s_i(h_l); h_l)$$

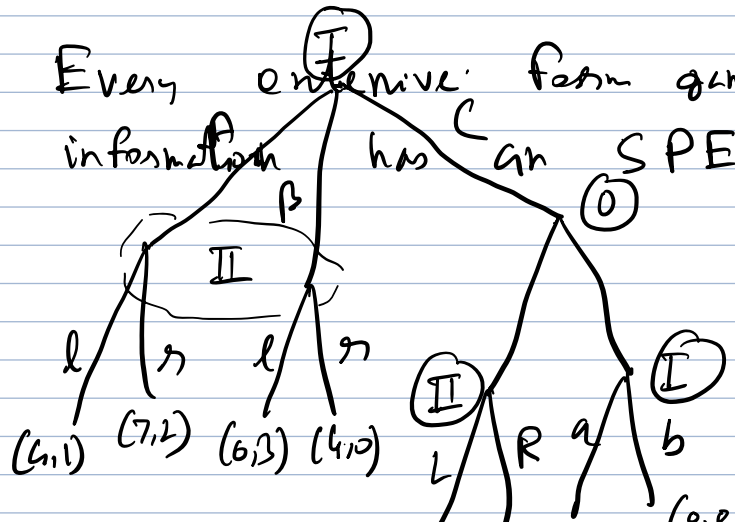
$$= \prod_{l=1}^{k} \beta_i(a_l; h_l) \prod_{l=k+1}^{m} (1) = \prod_{l=1}^{k} \beta_i(a_l; h_l)$$

$$= \prod_{j=1}^{k} \beta_i(a_j; h_j)$$

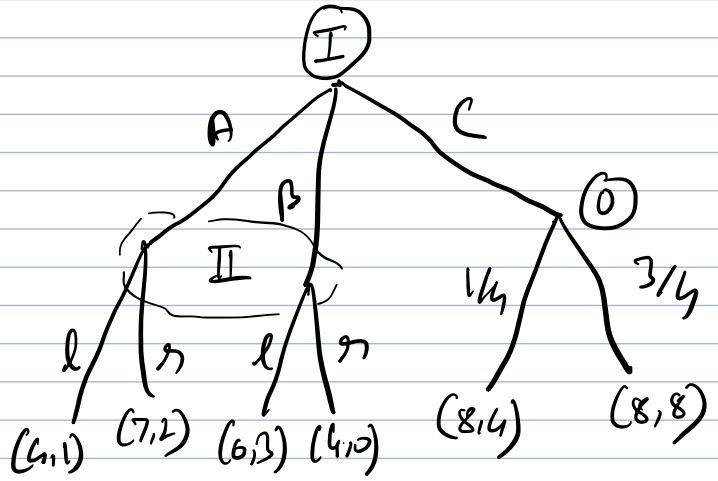
Therefore σ_i and β_i are outcome equivalent.

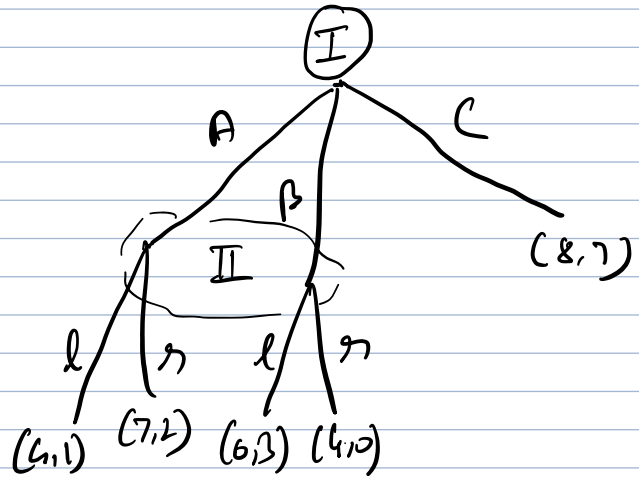
For the other part, a proof can be found in Solan's book.

Result Every extensive form game with imperfect information has an SPE.

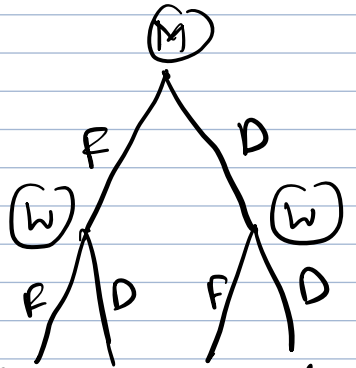


(0,0) (8,4) (8,8)





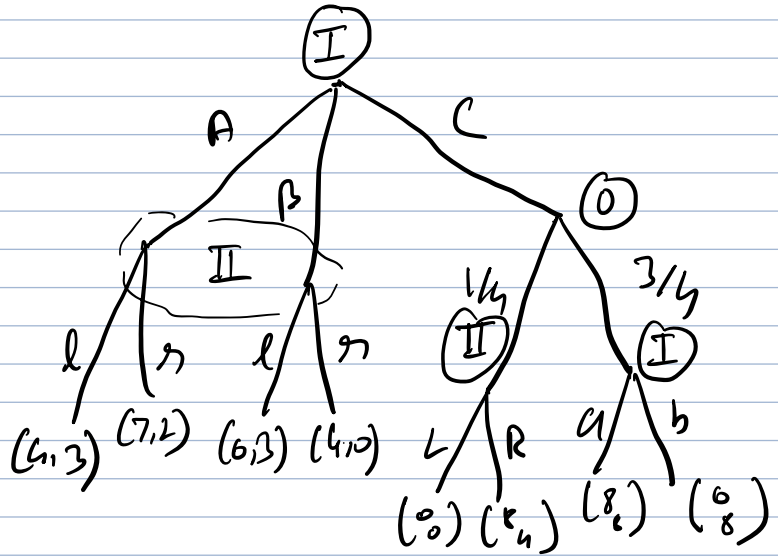
Backward induction over subgame. This will always result in an SPE.



(F, (F, F)) NB

(F, (F, D)) SPB

(2,1) (0,0) (0,0) (1,2)



(Ca, rR) is an SPE

Perfect Bayesian equilibrium, not in the syllabus.

- Given a Bayesian form game, can we get an extensive form game (equivalent)?

$$G = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$$

Assume that $S_i = \{s_i^1, \ldots, s_i^{m_i}\}$ for each i